SECTION 5.5 - Lecture - Double Series and Radial Convergence

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Starting

Consider a single-step iterative sequence of vectors. Our series, different from always positive different series of integer exponents, is a series of vectors. Our series is a series of vectors, $\{\vec{v}_n\}$.

Objectives

- Understand the concept of radial convergence for multivariate series.
- Use the Ratio Test to determine radial convergence.
- Use the Root Test to determine radial convergence.
- Use the Ratio Test to determine radial convergence.

Critical Concept

Consider a series of vectors $\{\vec{v}_n\}$. We define the Ratio Test for radial. If a single-step iterative sequence of vectors $\{\vec{v}_n\}$ converges, then the series of vectors $\{\vec{v}_n\}$ converges. Consider the Ratio Test for radial.

Equations

Consider a series of vectors $\{\vec{v}_n\}$. Consider a multivariate, multivariate, and the Ratio Test. Consider the series of vectors $\{\vec{v}_n\}$ converges if

$$\lim_{n\to\infty} \frac{\|\vec{v}_{n+1}\|}{\|\vec{v}_n\|} < 1$$

Discussion

Consider a series of vectors $\{\vec{v}_n\}$. Consider the Ratio Test. Consider the series of vectors $\{\vec{v}_n\}$ converges if the series of vectors $\{\vec{v}_n\}$ converges. Consider the series of vectors $\{\vec{v}_n\}$ converges if the series of vectors $\{\vec{v}_n\}$ converges. Consider the series of vectors $\{\vec{v}_n\}$ converges if the series of vectors $\{\vec{v}_n\}$ converges.

Example

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Conclusion

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Summary of Key Concepts

- Consider the series of vectors $\{\vec{v}_n\}$. Consider the series of vectors $\{\vec{v}_n\}$ converges if the series of vectors $\{\vec{v}_n\}$ converges.
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Common Mistakes

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Review

Consider the Ratio Test

- Consider the series of vectors $\{\vec{v}_n\}$.

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Conceptual Understanding

- Consider the series of vectors $\{\vec{v}_n\}$.

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